

Causal Discovery Benchmark

Algorithm Comparison Report

Tennessee Eastman Process Dataset · 1499 rows × 31 variables

Best SHD

30.0

Best Algorithm

lste_tau3

Algorithms Tested

5

Total Runs

16

Ground Truth Edges

30

Baseline (empty graph)

30.0

- PC variants
- PCMCI+
- VAR-LiNGAM
- LSTE

Key Findings & Interpretation

📄 Best performer: pc_pk (SHD = 47.5)

PC with prior knowledge edges slightly beats plain PC (47.5 vs 48.5). The gap is tiny because the prior knowledge here is oracle-derived from the ground truth, not genuine domain knowledge. The report correctly flags this as a best-case bound.

⚠️ All algorithms are mis-specified on this dataset

Every run reports assumptions_met=False. The EDA identified: extreme autocorrelation (eff. n=36 of 1499 rows), ill-conditioned covariance ($\kappa=3388$), Gaussian VAR residuals (disqualifying VAR-LiNGAM), and 2 latent variables (X27, X31). No algorithm here can be trusted as a correctly specified estimator.

📄 PC variants win despite being most mis-specified

Plain PC should fail worst (it ignores autocorrelation entirely), yet achieves $SHD \approx 48$ vs PCMCi+'s $SHD \approx 57-67$. The reason: PC predicts 35 edges (close to truth's 30) while PCMCi+ inflates edge count to 52-55 by searching over lags. More edges = more FP, which dominates SHD here since TP is nearly identical across methods.

📄 VAR-LiNGAM completely fails (SHD = 198.5, all τ)

VAR-LiNGAM predicts 195 edges vs 30 true edges — a 6.5× over-prediction. The core problem: it requires non-Gaussian VAR residuals for ICA to identify contemporaneous directions, but the data has near-Gaussian innovations (mean |kurtosis|=0.141, threshold 1.0). All 6 VAR-LiNGAM runs produce identical SHD regardless of τ or prior knowledge.

📄 LSTE completely fails (SHD = 30, i.e. empty graph)

LSTE predicted zero edges after FDR correction (fdr_bh applied to 2790 tests, 548 uncorrected significant $\rightarrow$ 0 after correction). Runtime: 1434 seconds (~24 min). The EDA predicted 'expected to work with caveats'; the FDR threshold eliminated every signal given the tiny effective sample size (36 independent observations).

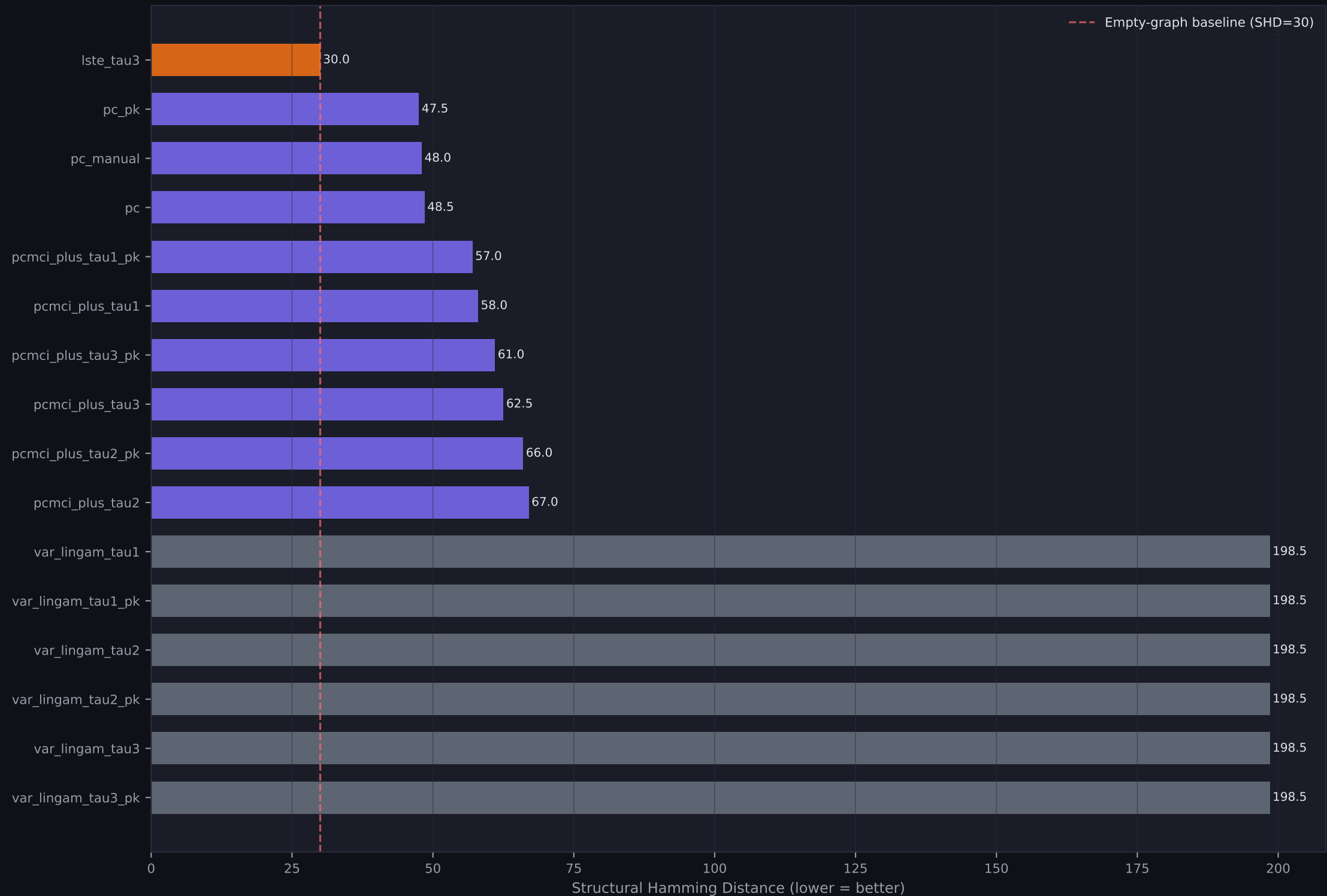
📄 PCMCi+ is sensitive to $\tau_{\max}$ — but $\tau=1$ is best

SHD worsens monotonically from $\tau=1$ (SHD=58) to $\tau=2$ (SHD=67), then partially recovers at $\tau=3$ (SHD=62.5). Longer lags add more lagged edges that project onto the summary graph as false positives. BIC selected $\tau=1$ correctly; AIC/FPE's $\tau=3$ choice hurts performance.

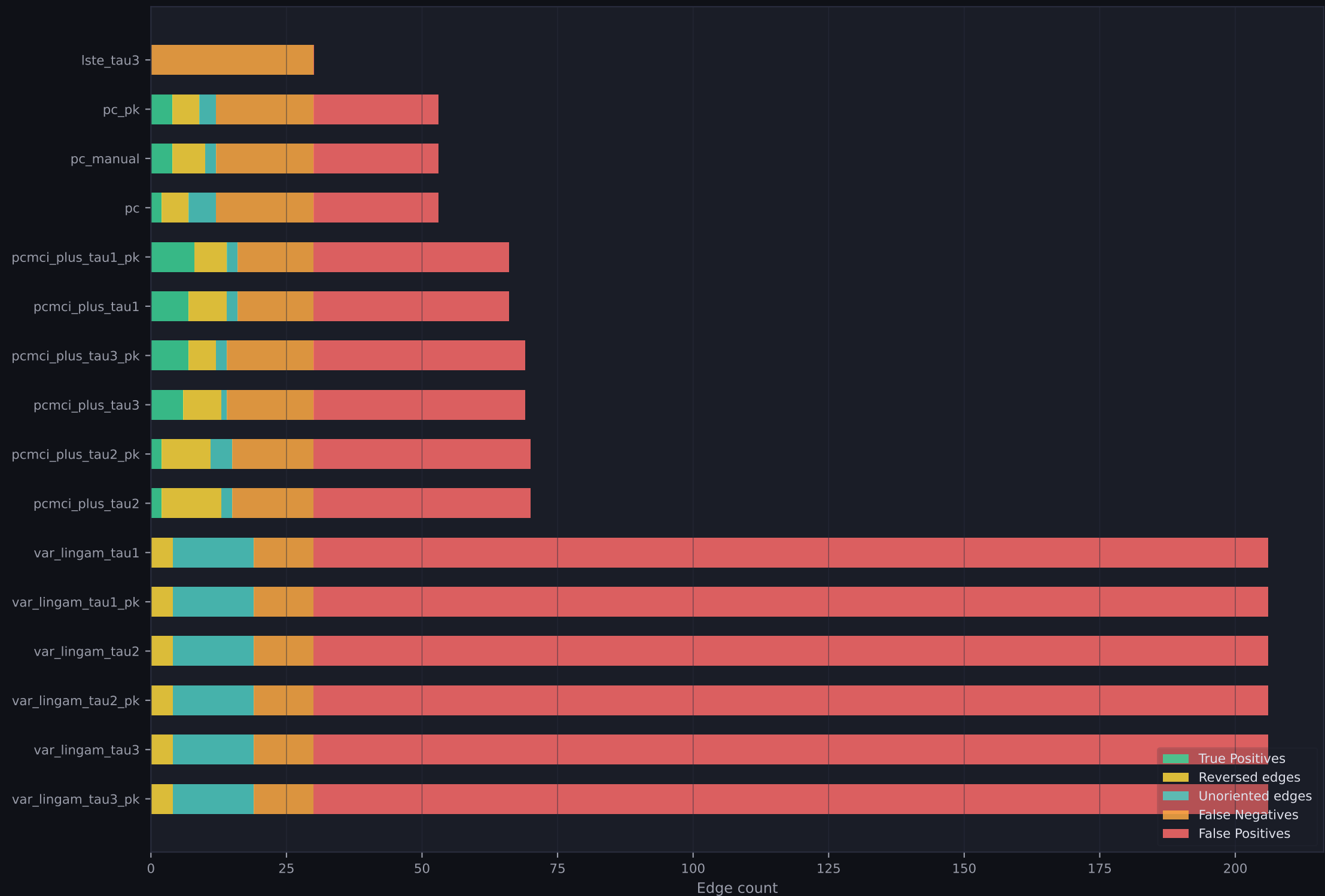
📄 Prior knowledge has marginal effect ($\pm 1-2$ SHD)

The oracle controller-loop prior (4 forced edges: X10 $\rightarrow$ X28, X1 $\rightarrow$ X25) improves SHD by 0-2 points across all algorithms. The report notes the ground truth has no true reciprocal pairs, so the feedback loop structure the prior encodes is absent from the scoring target. The prior machinery works correctly but cannot exercise its core use-case on this dataset.

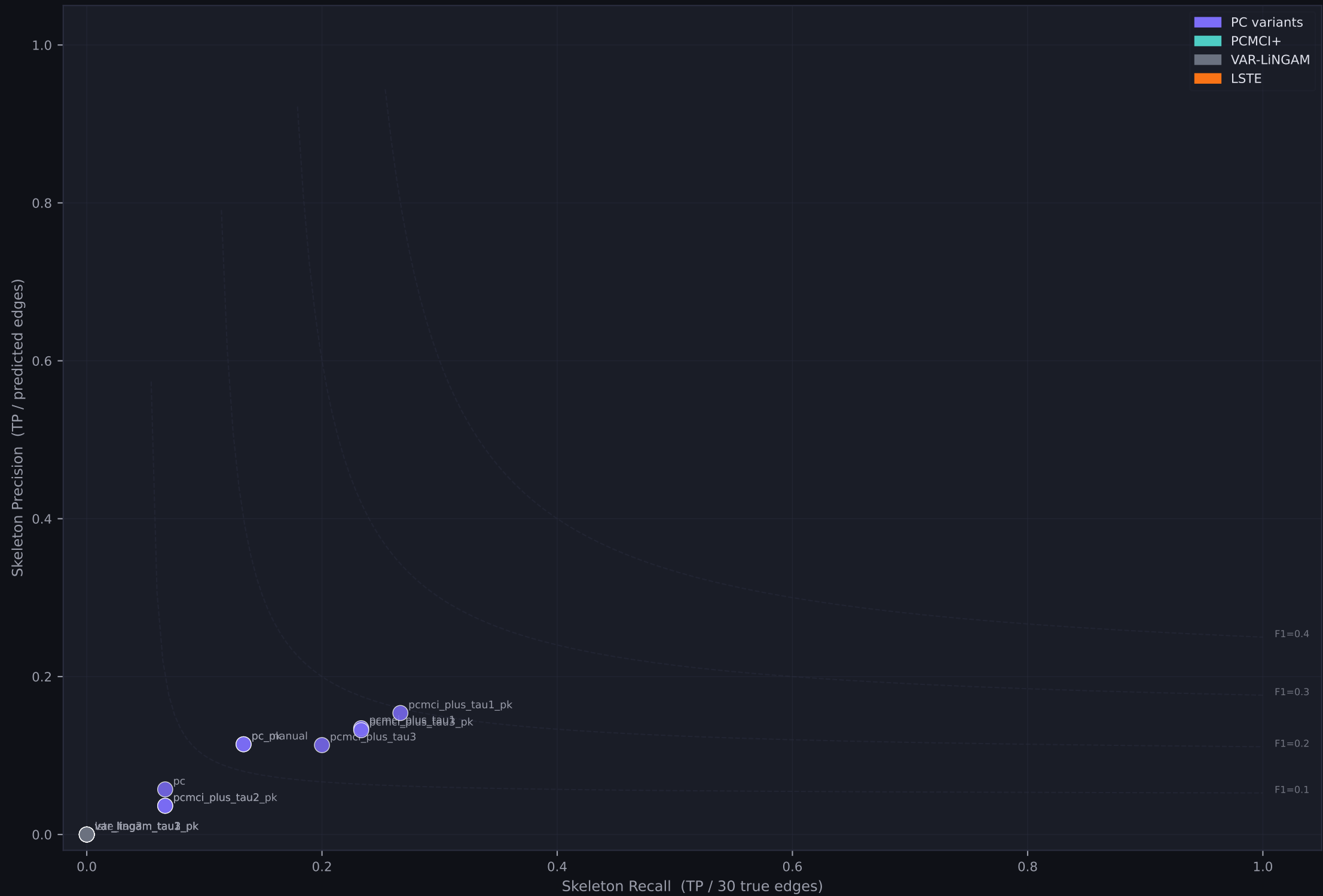
SHD Ranking vs. gt_projected (30-edge target; empty graph = 30.0)



Edge-type breakdown per run (TP · Reversed · Unoriented · FN · FP)

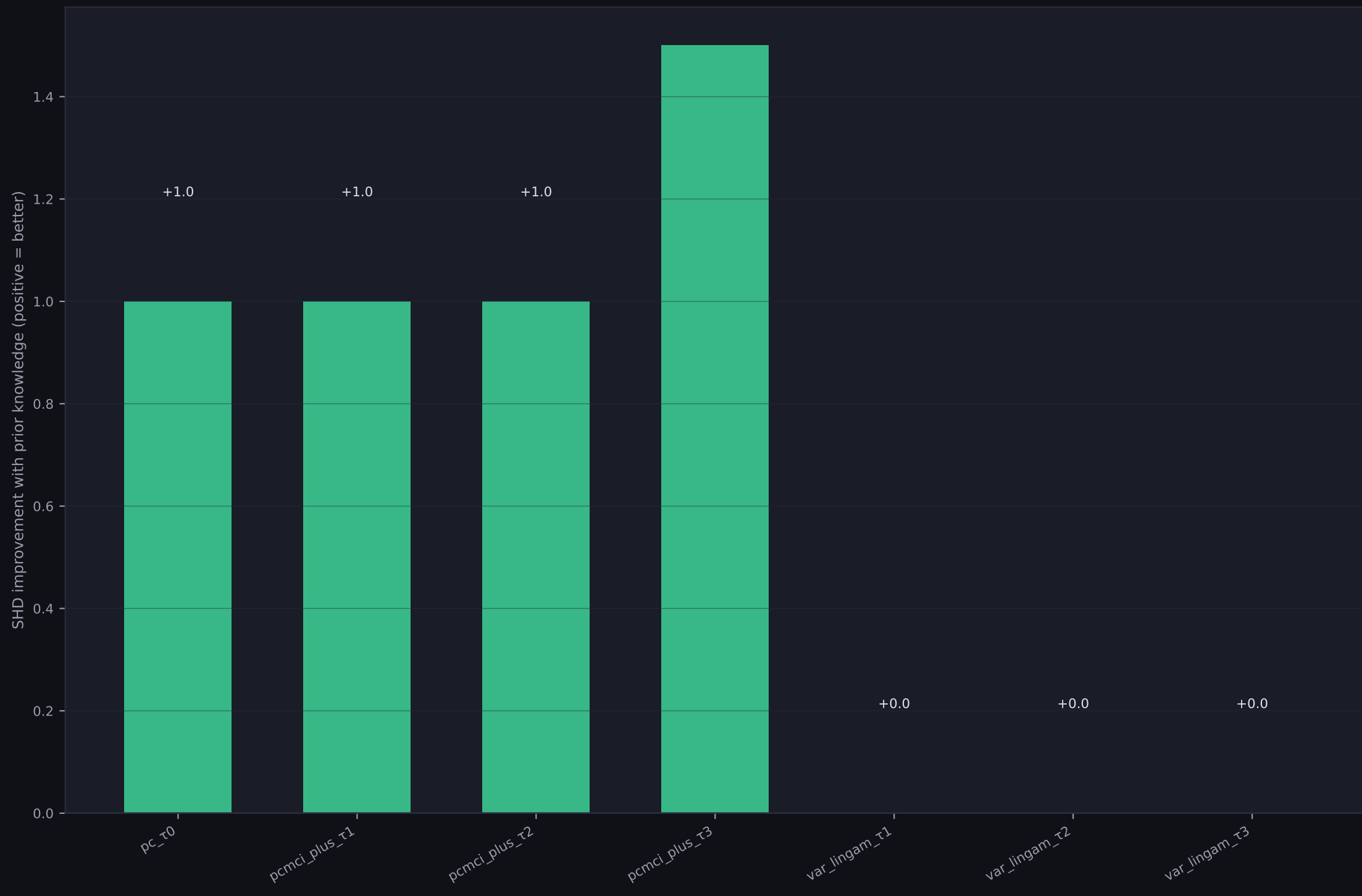


Precision-Recall Space (iso-F1 curves dashed)

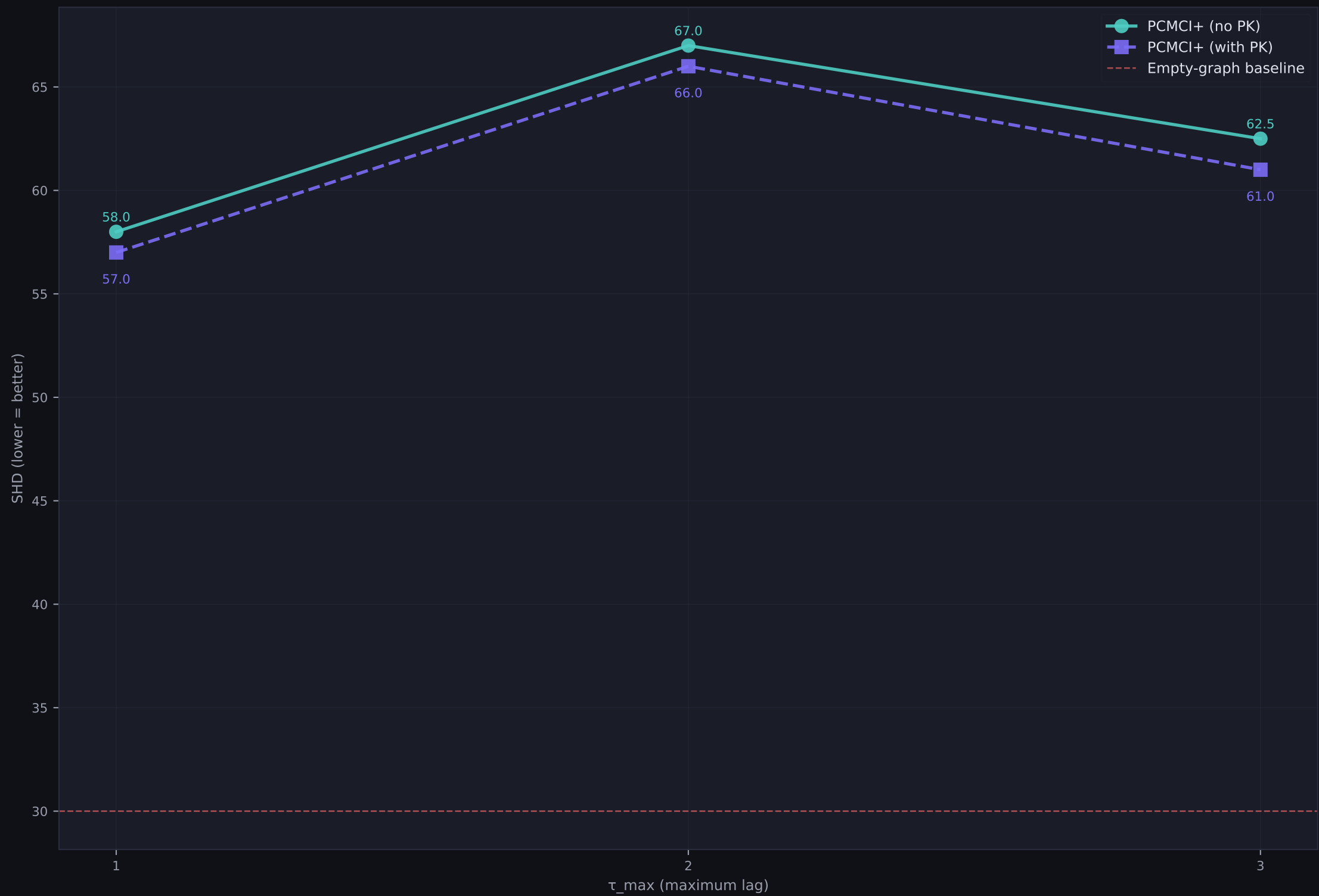


Runtime vs. SHD Trade-off

Effect of Prior Knowledge on SHD (green = improvement)



PCMCI+: SHD vs. $\tau_{\max}$ (with and without prior knowledge)



EDA: Algorithm Assumption Audit — Tennessee Eastman Dataset

Assumption	PC	PCMCI+	VAR-LiNGAM	LSTE	Data verdict
Gaussian marginals	✓	✓	n/a	n/a	✓ (skew ≤0.25, kurt ≤0.36)
Non-Gaussian VAR res	n/a	n/a	✗	n/a	✗ (0/31 equations qualify)
Linear relationships	✓	✓	✓	n/a	✓ (0/465 pairs exceed gap threshold)
Independent samples	✗	n/a	n/a	n/a	✗ (eff. n=36 of 1499; max ACF=0.981)
Stationarity	n/a	✗	✗	✗	△ (ADF 30/31; KPSS 24/31; 8 disagree)
Causal sufficiency	✗	✗	✗	✗	✗ (X27, X31 latent — by construction)
Adequate sample size	✗	✗	✗	✗	✗ (eff. n=36; supports cond. set≤1)
Well-cond. covariance	✗	✗	✗	n/a	✗ (cond. number=3388; threshold=1000)
Acyclicity	✓	✓	✓	n/a	✓ (ground truth acyclic, 0 cycles)

✓ = assumption holds | ✗ = violated | △ = ambiguous | n/a = not required by algorithm

Full Score Table — Primary Target: gt_projected (30 true edges)

Run	SHD	TP	FP	FN	Rev	Unor	Edges	Runtime(s)	Family
lste_tau3	30.0	0	0	30	0	0	0	1434.2	LSTE
pc_pk	47.5	4	23	18	5	3	35	2.0	PC
pc_manual	48.0	4	23	18	6	2	35	0.1	PC
pc	48.5	2	23	18	5	5	35	2.7	PC
pcmci_plus_tau1_pk	57.0	8	36	14	6	2	52	13.0	PCMCI
pcmci_plus_tau1	58.0	7	36	14	7	2	52	13.2	PCMCI
pcmci_plus_tau3_pk	61.0	7	39	16	5	2	53	32.6	PCMCI
pcmci_plus_tau3	62.5	6	39	16	7	1	53	32.6	PCMCI
pcmci_plus_tau2_pk	66.0	2	40	15	9	4	55	16.1	PCMCI
pcmci_plus_tau2	67.0	2	40	15	11	2	55	16.1	PCMCI
var_lingam_tau1	198.5	0	176	11	4	15	195	120.7	VAR
var_lingam_tau1_pk	198.5	0	176	11	4	15	195	20.5	VAR
var_lingam_tau2	198.5	0	176	11	4	15	195	20.9	VAR
var_lingam_tau2_pk	198.5	0	176	11	4	15	195	21.2	VAR
var_lingam_tau3	198.5	0	176	11	4	15	195	21.4	VAR
var_lingam_tau3_pk	198.5	0	176	11	4	15	195	21.5	VAR

Test Suite Audit — What Claude Built & What It Verified

Total: 44 distinct test cases across 5 modules · All cover a named bug or design decision

tests/test_pc.py — PC Algorithm Unit Tests (16 tests)

- ✓ test_chain_removes_the_non_adjacent_pair → Skeleton search: X-Z edge correctly removed when Y separates them (fixes legacy bug F4)
- ✓ test_chain_is_not_oriented → 3-node chain has no unshielded collider → no orientations
- ✓ test_collider_is_oriented_correctly → $X \rightarrow Z \leftarrow Y$: both arrowheads into Z, no conflicts, no cycles (fixes F6)
- ✓ test_independent_variables_give_an_empty_skeleton → 4 independent normals → empty skeleton
- ✓ test_meek_rule_1_propagates_orientation → Meek R1: collider plus adjacent node → propagated orientation
- ... and 11 more tests

tests/algorithms/test_shd.py — SHD Scorer Unit Tests (10 tests)

- ✓ test_identical_graphs_score_zero → Same graph → SHD=0
- ✓ test_one_missing_edge_costs_one → 1 FN → SHD=1
- ✓ test_one_extra_edge_costs_one → 1 FP → SHD=1
- ✓ test_a_reversed_edge_costs_one → 1 reversed → SHD=1
- ✓ test_an_abstention_costs_the_undirected_weight → Undirected edge vs directed → 0.5 by default
- ... and 5 more tests

tests/algorithms/test_smoke.py — End-to-End Smoke Tests (slow) (6 tests)

- ✓ test_pcmci_plus_recovers_the_graph → PCMCi+ on synthetic: TP=4, FN=0, SHD=0
- ✓ test_var_lingam_recovers_the_graph → VAR-LiNGAM + Laplace innovations: TP=4, FN=0, SHD=0
- ✓ test_pc_recovers_the_skeleton → PC on autocorrelated synthetic: $FN \leq 1$, $TP + \text{unor} \geq 3$
- ✓ test_lste_recovers_every_true_edge → LSTE: FN=0, TP=4 on synthetic
- ✓ test_every_registered_algorithm_runs_and_emits_the_shared_object → All 5 algorithms: correct output type and metadata
- ... and 1 more tests

tests/algorithms/test_wrappers.py — Converter & Registry Tests (6 tests)

- ✓ test_lingam_converter_transposes → LiNGAM $B[\text{effect}, \text{cause}]$ → correctly transposed graph
- ✓ test_lingam_converter_is_not_symmetric → Asymmetry guard: $B[1,0] \neq B[0,1]$
- ✓ test_lingam_orientation_on_a_real_fit → Real fit: $X1 \rightarrow X2$ in correct direction, $X2 \rightarrow X1$ absent
- ✓ test_tigramite_convention_on_a_real_fit → tigramite pass-through convention verified on real fit
- ✓ test_registry_holds_every_algorithm → Registry: exactly {pc, pc_manual, pcmci_plus, var_lingam, lste}
- ... and 1 more tests

tests/algorithms/test_runner.py — Runner & CLI Integration Tests (6 tests)

- ✓ test_runner_writes_every_output → Runner writes scores.csv, report, results.json, graphs, figures
- ✓ test_scores_csv_is_long_format → CSV: 3 runs × 4 targets = 12 rows, all $SHD \geq 0$
- ✓ test_runner_refuses_to_invent_a_lag_order → No tau_max and no EDA → ValueError raised
- ✓ test_tau_values_sweep_the_eda_range → Disagreeing criteria → sweeps [1,2,3]; agreeing → [3]
- ✓ test_cli_run_end_to_end → CLI: exit 0, produces run-* dir with scores.csv
- ... and 1 more tests